NEXUS

Causal Intelligence Platform — Risk Report

Repository: https://github.com/psf/requests

Analysis Window: 2023-01-01 to present

Generated: 2026-05-17T04:10:33

Total DPRs: 15

CTG Edges: 30

Decay Alerts: 5

Executive Summary

This report analyzes 15 Decision Provenance Records (DPRs) across the https://github.com/psf/requests codebase. **0 DPRs** have critical risk scores (>=70), **0 assumptions** are actively decaying, and **0 items** require immediate remediation (P0).

Risk Scores — Top 10

DPR	Title	Component	Risk	Blast	Decay	Active?
DPR-001	Core HTTP Client Delegation to	Core HTTP Client	62	100.0	50.0	no
DPR-003	Standardized CA Certificate Ve	Security (TLS/SSL)	62	100.0	50.0	no
DPR-002	Introduction of a Session Obje	Session Management	49	75.0	25.0	no
DPR-004	Explicit Python 3.10+ Requirem	Platform Compatibility	49	75.0	25.0	no
DPR-006	Centralized Exception Hierarch	Error Handling	49	75.0	25.0	no
DPR-005	Character Encoding Detection P	Content Decoding	48	50.0	50.0	no
DPR-007	pyproject.toml for Build Syste	Project Build & Packaging	41	50.0	25.0	no
DPR-012	`CaseInsensitiveDict` for HTTP	HTTP Header Handling	29	75.0	25.0	no
DPR-013	Extensible Authentication Syst	Authentication	29	75.0	25.0	no
DPR-015	Browser-Style Cookie Managemen	Cookie Management	29	75.0	25.0	no

Active Decay Alerts

[MONITORING] **DPR-001**: `urllib3` is a robust, well-tested, and actively maintained library suitable for

Evidence: No explicit evidence of decay yet, but the risk stems from `urllib3` being an external dependency. Potential evidence wo

[MONITORING] **DPR-001**: The performance and feature set provided by `urllib3` are sufficient for the div

Evidence: No explicit evidence of decay yet. Potential evidence would include: `Requests` users reporting performance regressions

[MONITORING] **DPR-003**: `certifi` provides a reliable, widely trusted, and regularly updated certificate

Evidence: No explicit evidence of decay yet. Potential evidence would include: `certifi` releases falling significantly behind maj

[MONITORING] **DPR-005**: `charset_normalizer` provides superior or equivalent performance and accuracy to

Evidence: No explicit evidence of decay yet. Potential evidence would include: a noticeable increase in user reports of garbled te

[MONITORING] **DPR-001**: The performance and feature set provided by `urllib3` are sufficient for the div

Evidence: This assumption is at risk if `Requests`'s strategic direction or user demands outpace `urllib3`'s development. For inst

Remediation Backlog

[P1] **DPR-001**: Core HTTP Client Delegation to urllib3 — Risk: 62

- Add blast radius monitoring and alerts

[P1] **DPR-003**: Standardized CA Certificate Verification via certifi — Risk: 62

- Add blast radius monitoring and alerts

[P2] **DPR-002**: Introduction of a Session Object for State Persistence — Risk: 49

- Monitor and reassess next cycle

[P2] **DPR-004**: Explicit Python 3.10+ Requirement — Risk: 49

- Monitor and reassess next cycle

[P2] **DPR-006**: Centralized Exception Hierarchy — Risk: 49

- Monitor and reassess next cycle

[P2] **DPR-005**: Character Encoding Detection Preference (charset_normalizer) — Risk: 48

- Monitor and reassess next cycle

[P2] **DPR-007**: pyproject.toml for Build System Standardization — Risk: 41

- Monitor and reassess next cycle

Knowledge Concentration — SPOF Alerts

Kenneth Reitz (initial design) → DPR-001: Core HTTP Client Delegation to urllib3 (66.7% ownership, critical blast radius)

Engineer 'Kenneth Reitz (initial design)' holds 66.7% of causal knowledge about Core HTTP Client Delegation to urllib3 (Core HTTP Client). Knowledge transfer recommended.

Kenneth Reitz → DPR-002: Introduction of a Session Object for State Persistence (66.7% ownership, high blast radius)

Engineer 'Kenneth Reitz' holds 66.7% of causal knowledge about Introduction of a Session Object for State Persistence (Session Management). Knowledge transfer recommended.

Kenneth Reitz → DPR-003: Standardized CA Certificate Verification via certifi (66.7% ownership, critical blast radius)

Engineer 'Kenneth Reitz' holds 66.7% of causal knowledge about Standardized CA Certificate Verification via certifi (Security (TLS/SSL)). Knowledge transfer recommended.

Kenneth Reitz → DPR-006: Centralized Exception Hierarchy (66.7% ownership, high blast radius)

Engineer 'Kenneth Reitz' holds 66.7% of causal knowledge about Centralized Exception Hierarchy (Error Handling). Knowledge transfer recommended.

Kenneth Reitz → DPR-012: `CaseInsensitiveDict` for HTTP Headers (66.7% ownership, high blast radius)

Engineer 'Kenneth Reitz' holds 66.7% of causal knowledge about `CaseInsensitiveDict` for HTTP Headers (HTTP Header Handling). Knowledge transfer recommended.

Kenneth Reitz → DPR-013: Extensible Authentication System (66.7% ownership, high blast radius)

Engineer 'Kenneth Reitz' holds 66.7% of causal knowledge about Extensible Authentication System (Authentication). Knowledge transfer recommended.

Kenneth Reitz → DPR-015: Browser-Style Cookie Management via `http.cookiejar` (66.7% ownership, high blast radius)

Engineer 'Kenneth Reitz' holds 66.7% of causal knowledge about Browser-Style Cookie Management via `http.cookiejar` (Cookie Management). Knowledge transfer recommended.

Counterfactual Simulation Summary

[EXTREME] What if Requests had chosen to implement its own low-level HTTP connection management instead of delegating to `urllib3`?

Affected: 6 DPRs across 6 components

[EXTREME] What if Requests had decided to rely on the system's CA certificates by default for SSL/TLS verification, instead of bundling `certifi`?

Affected: 6 DPRs across 6 components

[EXTREME] What if Requests had maintained compatibility with older Python versions (e.g., Python 3.6 or 3.7) instead of requiring Python 3.10+?

Affected: 6 DPRs across 6 components

[EXTREME] What if Requests had chosen to use only standard Python exceptions (e.g., `OSError`, `ConnectionError`) instead of establishing its own custom exception hierarchy?

Affected: 6 DPRs across 6 components

[LOW] What if Requests had used a standard Python dictionary for HTTP headers instead of a custom `CaseInsensitiveDict`?

Affected: 0 DPRs across 0 components